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1**Reflex arc components**

COMP

You should be able to: Diagram a reflex arc naming the receptor, sensory neuron, integrating center, motor neuron, and effector, and classify the reflex as monosynaptic or polysynaptic.

PICK ONE AND DO THAT ONE

☐ **A** Draw a spinal cord cross section and build one real reflex on it, your own choice, not the patellar. Label all five components in order, receptor, sensory neuron, integrating center, motor neuron, effector. Put the dorsal root and ventral root in their correct places and count the synapses inside the cord. Write monosynaptic or polysynaptic on the drawing. One line: what that synapse count buys and what it costs.

☐ **B** Draw one reflex arc, then cut it in four separate places, at the receptor, at the dorsal root, inside the cord, and at the ventral root. For each cut, draw what the person can still do and what they cannot, keeping sensation and movement separate. One line: which two cuts look the same from the outside, and the one test that would separate them.

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2**Muscle spindle and stretch reflex**

COMP

You should be able to: Explain how the muscle spindle detects stretch and trace the stretch reflex including reciprocal inhibition of the antagonist.

PICK ONE AND DO THAT ONE

☐ **A** Draw a muscle with its extrafusal fibers and one spindle drawn in parallel with them, plus the antagonist muscle beside it. Stretch the muscle on your drawing, then trace the circuit from the spindle afferent into the cord and back out, showing both the excitatory path to its own muscle and the inhibitory interneuron to the antagonist. One line: what the limb would do if that inhibitory interneuron were missing.

☐ **B** Draw the same spindle three times, at rest, during a sudden stretch, and during a shortening contraction with no gamma motor neuron activity. In each, show the spindle's own length and its firing rate. Add a fourth panel with gamma motor neurons firing alongside the alpha motor neurons. One line: what the gamma neurons are protecting the spindle from.

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3

COMP

Golgi tendon organ

You should be able to: Contrast the stimulus and the reflex response of the Golgi tendon organ with those of the muscle spindle.

PICK ONE AND DO THAT ONE

☐ **A** Draw one muscle with its tendon. Place the spindle and the Golgi tendon organ each in its correct physical position, one in parallel with the fibers and one in series at the tendon. Beside each, write the stimulus it actually detects and draw its reflex output back to that same muscle with a plus or a minus sign. One line: why something in series can report what a parallel structure cannot.

☐ **B** Draw a person holding a load with the elbow bent and still. Underneath, draw two firing rate traces on one shared time axis, the spindle afferent and the Golgi tendon organ afferent, as the load is slowly increased. Beside each, draw what its reflex is trying to make the muscle do. One line: which one wins if the load keeps climbing, and what you would see happen to the arm.

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4

COMP

Withdrawal and crossed extensor reflexes

You should be able to: Trace the flexor withdrawal reflex and the crossed extensor reflex and explain how the two together preserve balance.

PICK ONE AND DO THAT ONE

☐ **A** Draw a standing person from the waist down, both legs, who has just stepped on something sharp with the right foot. Put the cord in the middle and trace the circuit to all four muscle groups, right flexors, right extensors, left flexors, left extensors. Mark each excited or inhibited and draw an interneuron everywhere one is needed. One line: what the left leg is doing, and why it has to happen at the same moment.

☐ **B** Draw this response twice, once complete and once with the crossed extensor half deleted. Show the body position a fraction of a second later in each drawing. Then count the synapses along the shortest path in your complete version. One line: using your own count, say why this reflex cannot be monosynaptic.

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5

Ascending and descending pathways

COMP

You should be able to: Compare the dorsal column, spinothalamic, and corticospinal pathways by the information carried, the side of decussation, and the deficit produced by a lesion.

PICK ONE AND DO THAT ONE

☐ **A** Draw a tall diagram with spinal cord at the bottom, medulla in the middle, and cortex on top. Run three tracts up or down it, dorsal column, spinothalamic, and corticospinal. Mark where each one crosses the midline and put an arrow on each showing which way information travels. Beside each, write what it carries. One line: two of them cross at very different levels, say why that matters at the bedside.

☐ **B** Draw a spinal cord cross section and cut clean through the left half only. Below it, draw a body outline and shade what is lost on each side, using a different mark for fine touch and position sense, for pain and temperature, and for voluntary movement. One line: name the finding on your drawing that tells you the lesion is in the cord and not in the brain.

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6

Cerebrospinal fluid and the blood brain barrier

COMP

You should be able to: State the functions of cerebrospinal fluid and explain how the blood brain barrier selects what reaches neurons, including why lipid soluble drugs cross readily.

PICK ONE AND DO THAT ONE

☐ **A** Draw the brain in section with the ventricles, and trace cerebrospinal fluid with arrows all the way from where it is made to where it is reabsorbed. Label the choroid plexus and the arachnoid villi. Beside the drawing, write the total volume, about 150 mL, and the amount produced per day, about 500 mL. One line: what that turnover rate means for a patient with a blockage in the loop.

☐ **B** Draw a brain capillary in cross section beside a capillary from elsewhere in the body, same scale. Show the junctions between endothelial cells in each and the astrocyte feet on the brain one. Now send four things at the brain capillary, oxygen, glucose, a large plasma protein, and a lipid soluble anesthetic. Draw how each gets through or fails. One line: why the drug that crosses easiest is also the hardest to keep out.

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7**Reflex testing and reaction time**

COMP

You should be able to: Elicit and grade deep tendon reflexes and measure reaction time to distinguish a reflex response from a voluntary response by latency.

PICK ONE AND DO THAT ONE

☐ **A** Draw a horizontal grading scale for deep tendon reflexes from 0 to 4 with every grade labeled. Mark where your own measured reflexes fell and where a hyperreflexic patient would fall. Beside it, draw the arc you were actually testing with its synapses shown. One line: a 0 and a 4 point at damage in different places, say which is which and how you would tell.

☐ **B** Draw one time axis in milliseconds. Mark the stimulus, then plot two responses on it using your own numbers, your measured tendon reflex latency and your measured reaction time from the ruler drop. Under each, draw the path that signal took and count the structures it passed through. One line: what accounts for the gap between the two, judging from your own drawings.

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8**Sensory transduction**

COMP

You should be able to: Explain how a stimulus is converted into a receptor potential and how that graded potential becomes a train of action potentials in the sensory neuron.

PICK ONE AND DO THAT ONE

☐ **A** Draw one sensory neuron stretched across your space, from its receptor ending to a point well along the axon. Above it, draw the membrane potential recorded at three places, the receptor ending, the trigger zone, and further down the axon. Label which traces are graded and which are all or none, and mark threshold. One line: what happens to the information at the trigger zone that cannot be undone afterwards.

☐ **B** Draw three stimulus strengths as three bars of different height on a time axis, weak, medium, and strong. Under each, draw the receptor potential it produces, and under that, the train of action potentials leaving the trigger zone. Keep every spike exactly the same height. One line: since the spikes are all identical, name what actually carries the message that the third stimulus was the strongest.

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9

COMP

Receptor classification

You should be able to: Classify a receptor by stimulus type as a mechanoreceptor, chemoreceptor, thermoreceptor, photoreceptor, or nociceptor and give a body location for each.

PICK ONE AND DO THAT ONE

☐ **A** Draw a body outline and place one real receptor from each of the five classes on it, mechanoreceptor, chemoreceptor, thermoreceptor, photoreceptor, and nociceptor. Put each where it genuinely exists, inside the body as well as on the surface. Beside each, draw the stimulus arriving as an arrow and name the form of energy it carries. One line: which class is spread over the most locations, and what that says about its job.

☐ **B** Pick a moment, drinking something hot, walking into the cold, or one of your own. Draw the scene and mark every receptor class firing during it, with an arrow from each stimulus to its receptor. Some classes will appear more than once at different sites. One line: name one class that is not firing in your scene, and the change that would recruit it.

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10

COMP

Stimulus coding

You should be able to: Explain how the nervous system encodes modality, location, intensity, and duration and identify the labeled line principle.

PICK ONE AND DO THAT ONE

☐ **A** Draw four panels in a row, one each for modality, location, intensity, and duration. In each, draw the actual mechanism, a labeled line running to a specific cortical area, a receptive field on the skin, a firing rate, and a firing pattern held over time. Use spikes on a time axis wherever a rate is the answer. One line: which of the four still works if you stimulate the nerve trunk directly with an electrode.

☐ **B** Draw what happens when you press hard on your own closed eyelid. Trace the path from the pressed photoreceptors to the cortical area that receives them, and draw what the person actually perceives. Beside it, draw the same mechanical pressure applied to the back of the hand and what that produces. One line: the same stimulus gave two different sensations, say what decided which one.

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11**Receptive fields and acuity**

COMP

You should be able to: Relate receptive field size and lateral inhibition to two point discrimination and predict which body regions have the finest acuity.

PICK ONE AND DO THAT ONE

☐ **A** Draw two patches of skin side by side, a fingertip and the middle of the back, with the sensory neuron receptive fields on each drawn to scale as circles. Set the same pair of caliper points on both and show why one is felt as two points and the other as one. Write your own measured thresholds in millimeters on each patch. One line: what would have to change on the back to make it match the fingertip.

☐ **B** Draw three sensory neurons in a row receiving a stimulus that is strongest on the middle one. Add the inhibitory interneurons that connect them sideways. Then draw the firing rate of all three as bars, once before the lateral connections act and once after. One line: what the sharpened pattern lets the brain do that the raw pattern did not.

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12**Receptor adaptation**

COMP

You should be able to: Distinguish tonic from phasic receptors by their adaptation rate and match each to a stimulus the body needs to monitor continuously or only at onset.

PICK ONE AND DO THAT ONE

☐ **A** Draw one stimulus bar on a time axis, held perfectly steady from start to finish. Under it, on the same axis, draw two firing rate traces responding to that same stimulus, a tonic receptor and a phasic receptor. Show clearly where the phasic one falls silent while the stimulus is still on, and mark any response when it switches off. Name a real receptor for each trace. One line: what the phasic receptor is reporting that the tonic one is not.

☐ **B** Pick two things the body monitors, one it needs to know continuously and one it only needs at the moment of change. Choose your own, not the shirt on your skin. Draw the receptor you would assign to each and its firing over a long steady stimulus. Then swap the two assignments and draw what goes wrong in each case. One line: which swap is actually dangerous, and why.

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13**Somatosensory pathways**

COMP

You should be able to: Trace touch and pain information from receptor to cortex and explain why the somatosensory homunculus is disproportionate.

PICK ONE AND DO THAT ONE

☐ **A** Draw one diagram running from a fingertip up to the cortex with two tracts on it, one carrying fine touch and one carrying pain. Mark where each crosses the midline, where each synapses on the way, and where each arrives in the postcentral gyrus. Use two colors or two line styles. One line: name a level where a single lesion would knock out one of them and spare the other.

☐ **B** Draw a strip of postcentral gyrus in cross section and lay the homunculus along it, sizing each body part by the cortical area it gets rather than by its real size. Beside it, draw the same body parts at true relative size. Use your own two point discrimination data to decide what to enlarge. One line: given both drawings, state what cortical area is actually proportional to.

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14**Pain and its modulation**

COMP

You should be able to: Distinguish fast and slow pain by fiber type, explain referred pain by convergence, and describe how gate control and endogenous opioids reduce pain transmission.

PICK ONE AND DO THAT ONE

☐ **A** Draw one painful stimulus and the two fibers carrying it away, an A delta and a C fiber, drawn to show the difference in diameter and myelination. On a shared time axis, mark when each signal arrives and write the sensation each produces. Then add the dorsal horn gate, a large touch fiber, the inhibitory interneuron, and the projection neuron. One line: why rubbing a bruise helps, judging from your own circuit.

☐ **B** Draw a spinal cord segment with a visceral afferent from the heart and a somatic afferent from the left arm skin converging on the same projection neuron. Follow that one neuron up to the cortex and draw where the brain places the pain. Then draw an endogenous opioid acting somewhere on this circuit and mark whether it works before or after the synapse. One line: why the brain guesses skin and not heart.

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15**Tactile mapping and adaptation testing**

COMP

You should be able to: Measure two point discrimination at several body sites and show receptor adaptation and referred sensation experimentally.

PICK ONE AND DO THAT ONE

☐ **A** Draw a body outline and write your own measured two point threshold in millimeters at every site you tested. Before looking at anyone else's data, circle the site you predicted would be finest. Then draw a bar for each site ordered from finest to coarsest with the real numbers on them. One line: name a site where your measurement and your prediction disagreed, and one method error that could explain it.

☐ **B** Draw a time axis and plot your own adaptation data on it, how long the sensation took to fade for each stimulus you tested. Beside it, draw a body outline with your referred sensation result on it, marking where the stimulus was applied and where it was actually felt. One line: what the distance between those two marks tells you about where in the pathway the mistake is being made.

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16**Optics and accommodation**

COMP

You should be able to: Explain refraction by the cornea and lens and describe how accommodation and the pupillary reflex focus near and far images.

PICK ONE AND DO THAT ONE

☐ **A** Draw an eye in cross section twice at the same scale with light rays coming in. In the first, the object is far away. In the second, it is close. In each, draw the lens shape, the ciliary muscle, the zonule fibers taut or slack, and where the rays come to a focus. One line: name which of the two states has the ciliary muscle working, and why that catches most people out.

☐ **B** Draw the pupillary light reflex as a circuit, light into one eye, the afferent nerve, the brainstem, and the efferent nerve running back out to both irises. Show the muscle that contracts and draw both pupils before and after. Then move the light to the other eye and draw the result. One line: what a response missing in one eye only tells you about where the lesion sits.

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17**Phototransduction**

COMP

You should be able to: Trace phototransduction from photon absorption by retinal through the change in cyclic GMP to the decrease in glutamate release and explain why photoreceptors hyperpolarize to light.

PICK ONE AND DO THAT ONE

☐ **A** Draw a photoreceptor twice, side by side, once in darkness and once in bright light. In each, show the cyclic GMP level, whether the cation channels are open or closed, the membrane potential, and how much glutamate is leaving the base of the cell. One line: name the state in which the cell releases the most transmitter, and say why that is the opposite of what you would expect.

☐ **B** Draw the cascade as a chain of labeled boxes from photon absorption through retinal, the pigment, the G protein, the enzyme, cyclic GMP, and the channel, with arrows between every box. On each arrow, mark whether the next thing goes up or down. Then break one box of your choosing and write what the cell now does when light hits it.

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18**Rods cones and visual processing**

COMP

You should be able to: Compare rods and cones by sensitivity and acuity and trace the visual pathway to predict the field deficit produced by a lesion at the optic nerve or the optic chiasm.

PICK ONE AND DO THAT ONE

☐ **A** Draw a patch of retina at the fovea and a patch from the far periphery, side by side, each showing photoreceptors, bipolar cells, and one ganglion cell. Make the convergence ratio visible by how many photoreceptors feed that one ganglion cell. Beside each, write which receptor type dominates, its sensitivity, and its acuity. One line: why the same wiring cannot give you both sensitivity and acuity at once.

☐ **B** Draw the visual pathway from both eyes through the chiasm to the cortex, marking which fibers cross and which do not. Then cut it in two places, once at the left optic nerve and once through the middle of the chiasm. For each cut, draw the two visual fields as a pair of circles and shade what is lost. One line: which lesion the patient notices first, and why.

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19

Refractive errors and clinical vision

COMP

You should be able to: Explain myopia, hyperopia, presbyopia, and astigmatism in terms of focal point and eyeball or lens shape and state the corrective lens for each.

PICK ONE AND DO THAT ONE

☐ **A** Draw four eyes in a row with parallel rays entering each, normal, myopic, hyperopic, and astigmatic. In each, mark where the rays come to a focus relative to the retina and mark whether the eyeball shape or the corneal surface is the problem. Then draw the corrective lens in front of each, bending the rays back onto the retina. One line: why the astigmatic eye needs a differently shaped lens than the other two.

☐ **B** Draw one eye reading a page at 25 cm at age 20 and the same eye at age 60, showing the lens shape and where the image falls in each. Beside them, draw a hyperopic 20 year old reading the same page. Both people end up holding it further away. One line: name the one structure that differs between the two cases, and a test that would tell them apart.

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20

Auditory transduction

COMP

You should be able to: Trace sound from the tympanic membrane through the ossicles and cochlear fluid to hair cell bending and auditory nerve firing and explain how pitch and loudness are coded.

PICK ONE AND DO THAT ONE

☐ **A** Draw the path of one sound from air to auditory nerve, tympanic membrane, the three ossicles named in order, oval window, cochlear fluid, basilar membrane, hair cell, nerve. Draw the round window and show what it is doing. Enlarge one hair cell and draw its stereocilia bending and the channel that opens. One line: name what the ossicles are correcting for at the boundary between air and fluid.

☐ **B** Draw the basilar membrane unrolled as one long strip, narrow and stiff at the end nearest the oval window and wide and floppy at the far end. Mark where a high pitched sound peaks and where a low one peaks. Then draw two hair cell firing traces for the same pitch at two loudness levels. One line: pitch and loudness are coded two different ways, say which one is place and which is rate.

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21**Conductive and sensorineural hearing loss**

COMP

You should be able to: Distinguish conductive from sensorineural hearing loss by the site of the lesion and by the expected Weber and Rinne results.

PICK ONE AND DO THAT ONE

☐ **A** Draw the ear in one cross section from the canal through to the auditory nerve, with outer, middle, and inner ear marked off. Put an X at a site that would cause conductive loss and a different X at a site that would cause sensorineural loss. Beside each X, draw the tuning fork on the mastoid and beside the ear, and mark which position is heard louder. One line: name what bone conduction bypasses.

☐ **B** Draw a head from the front three times with a tuning fork on the midline of the skull in each, once with normal hearing, once with a conductive loss on the right, and once with a sensorineural loss on the right. Draw an arrow from the fork to the ear where the sound is heard louder in each. One line: those arrows point opposite ways, say what it costs you to run Weber alone.

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22**Equilibrium**

COMP

You should be able to: Explain how the semicircular canals detect angular acceleration and how the utricle and saccule detect linear acceleration and head position.

PICK ONE AND DO THAT ONE

☐ **A** Draw one semicircular canal with its cupula and one macula with its otolith membrane and otoconia, side by side and both enlarged. On each, draw the head movement it detects as an arrow and show what physically pushes on the hair cells in that case. One line: name a head movement that bends the cupula but leaves the otoliths sitting where they are.

☐ **B** Draw the endolymph and cupula inside one horizontal canal at four moments, sitting still, at the start of a spin, during a long steady spin at constant speed, and the instant you stop. Show the cupula position in every panel. One line: using your fourth panel, say why you still feel like you are turning after you have stopped.

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23**Taste and smell**

COMP

You should be able to: Compare gustatory and olfactory transduction and explain why olfactory input reaches the limbic system without a thalamic relay.

PICK ONE AND DO THAT ONE

☐ **A** Draw a taste bud with one taste cell enlarged beside an olfactory epithelium with one olfactory receptor neuron enlarged, at the same scale. On each, show the stimulus arriving, what it binds to or passes through, and how the cell signals the next neuron along. Mark which of your two cells is itself a neuron. One line: name a taste that works by an ion entering the cell directly rather than by a receptor protein.

☐ **B** Draw both pathways from receptor to cortex on one page, taste on one side and smell on the other. Put the thalamus in the middle and route each pathway correctly, through it or around it. Mark where the smell pathway reaches the limbic system. One line: why a smell can drop you into a memory before you have even named the smell.

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24**Vision testing**

COMP

You should be able to: Perform visual acuity, blind spot, accommodation, and color vision testing and interpret each result.

PICK ONE AND DO THAT ONE

☐ **A** Draw four panels in a row, one for each test you ran, acuity, blind spot, accommodation, and color vision. In each panel, draw the setup and your own result, with real numbers where you have them, including your acuity fraction and your near point in centimeters. One line: name the test whose result depends most on the subject being honest, and what that means for trusting the number.

☐ **B** Draw the eye in cross section and mark the optic disc and the fovea in their correct positions relative to each other. Draw rays from the target in your blind spot experiment to work out where the spot falls in the visual field and on which side. Then do the same for the other eye. One line: why you never notice the blind spot in ordinary life.

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25

Hearing and equilibrium testing

COMP

You should be able to: Perform Weber and Rinne tuning fork tests and a balance test and interpret the findings.

PICK ONE AND DO THAT ONE

☐ **A** Draw the Weber and Rinne procedures as you actually performed them, with the fork position marked on a head for each and your own subject's result written beside it. Then add your balance test, drawing the subject's stance and marking how the sway changed when the eyes closed. One line: name which system takes over when vision is removed, judging from your own sway result.

☐ **B** Draw a branching chart of boxes and arrows that starts at a Weber result and ends at a named finding, with Rinne as the second branch. Fill in every ending, normal, conductive on one side, sensorineural on one side. Then run your own subject's results through it and mark the path they took. One line: name an ending your chart cannot reach, and what you would need to add to reach it.

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Dr. Sharilyn Rennie